

R Programming Tutorial

#08

Correlation & Regression

#lectureSEVEN

Prepared by: Chan Weng Howe

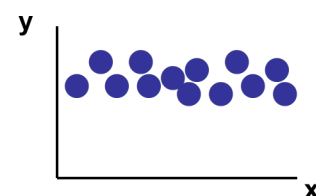
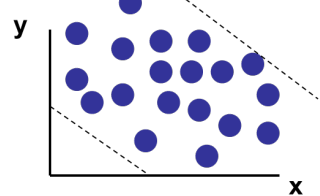
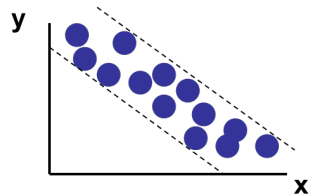
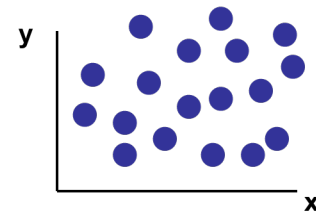
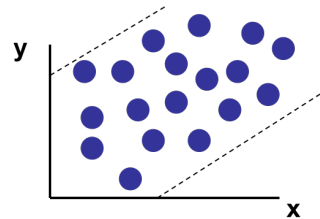
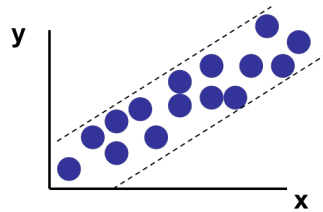
#08-1

CORRELATION & REGRESSION

REVISION

Correlation Analysis

- Measure the **strength** of relationship between **two** variables.

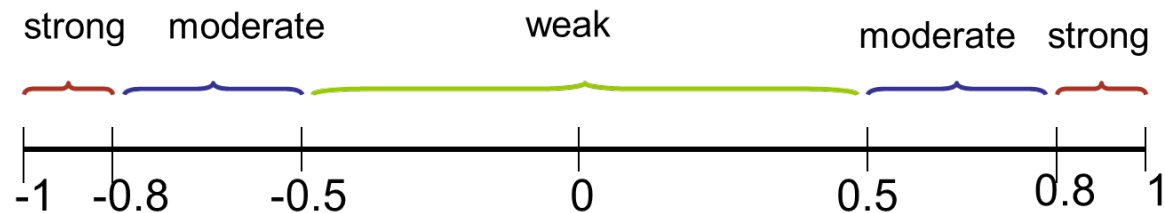


Strong
relationship

Weak
relationship

No
relationship

Strength of
correlation
coefficient (r)



#08-2

CORRELATION ANALYSIS IN R

PEARSON CORRELATION

Correlation Analysis

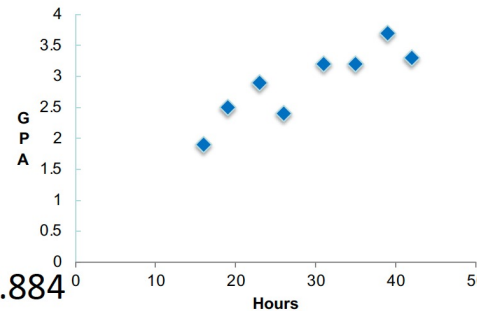
Pearson Correlation

REFER to:
lectureSEVEN #01

POSITIVE CORRELATION (slide 18)

Positive Correlation

- What is the relationship between the number of hours spent per week studying and GPA?



Students	Study Hours	GPA
S1	42	3.3
S2	23	2.9
S3	31	3.2
S4	35	3.2
S5	16	1.9
S6	26	2.4
S7	39	3.7
S8	19	2.5

It can be seen that the GPA increases as the hours increases. A scatter plot and correlation analysis of the data indicates that there is positive relationship between the number of hours spent per week studying and GPA.

Correlation Analysis

Pearson Correlation

- Correlation efficient (r) values can be calculated in R via **cor()** function.
- How to use? **cor(x,y)**
 - The default method used is **Pearson's**
- Example data:

Students	Study Hours	GPA
S1	42	3.3
S2	23	2.9
S3	31	3.2
S4	35	3.2
S5	16	1.9
S6	26	2.4
S7	39	3.7
S8	19	2.5

```

1  x <- c(42,23,31,35,16,26,39,19)
2  y <- c(3.3, 2.9, 3.2, 3.2, 1.9,
3     2.4, 3.7, 2.5)
4
5     #calculate corr. coefficient
6     cor(x,y)
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22

```

```

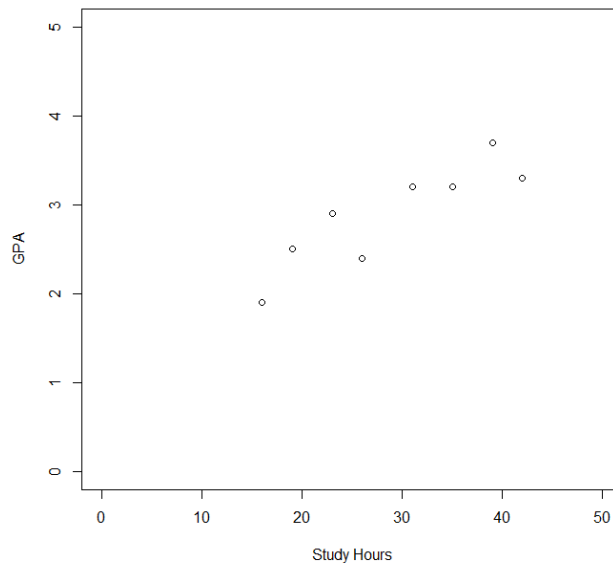
> cor(x,y)
[1] 0.8837091

```

Correlation Analysis

Pearson Correlation

- How we can view the data in graph?
via **Scatter plot**
- Use simple **plot()** function in R.



```

1  x <- c(42,23,31,35,16,26,39,19)
2  y <- c(3.3, 2.9, 3.2, 3.2, 1.9,
3    2.4, 3.7, 2.5)
4
5      #calculate corr. coefficient
6      cor(x,y)
7
8      plot(x,y, xlim=c(0,50),
9           ylim=c(0,5), xlab="Study
10     Hours", ylab="GPA")
11
12
13
14
15
16
17
18
19
20
21
22
  
```

Correlation Analysis

Pearson Correlation

REFER to:

lectureSEVEN #01

EXERCISE #2 (slide 30)

Exercise #2

- The data represent x = score on a measure of test anxiety and y = exam score for a sample of $n = 9$ students:

x	23	14	14	0	17	20	20	15	21
y	43	59	48	77	50	52	46	51	51

Higher values for x indicate higher levels of anxiety.

Correlation Analysis

Pearson Correlation

x	23	14	14	0	17	20	20	15	21
y	43	59	48	77	50	52	46	51	51

The data represent x=score on a measure of test anxiety and y=exam score for a sample of n=9 students:

- Construct a scatter plot, and comment on the features of the plot.
- Does there appear to be a linear relationship between the two variables? How would you characterize the relationship?
- Compute the value of the correlation coefficient. Is the value of r consistent with your answer to part (b)?
- Is it reasonable to conclude that test anxiety caused poor exam performance? Explain.

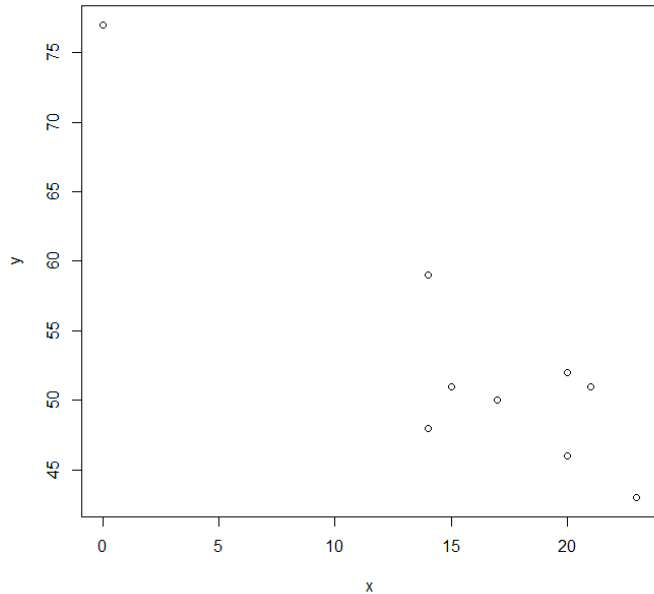
```

1  x<- c(23,14,14,0,17,20,20,15,21)
2  y<- c(43,59,48,77,50,52,46,51,51)
3
4  plot(x,y)
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
  
```

Correlation Analysis

Pearson Correlation

The plotted graph in R



```
1 x<- c(23,14,14,0,17,20,20,15,21)
2 y<- c(43,59,48,77,50,52,46,51,51)
```

```
3
4 plot(x,y)
```

```
5
6 cor(x,y)
```

```
7
```

```
8
```

```
9
```

```
10
```

```
11
```

```
12
```

```
13
```

```
14
```

```
15
```

```
16
```

```
17
```

```
18
```

```
19
```

```
20
```

```
21
```

```
22
```

```
> cor(x,y)
[1] -0.9124331
```

#08-3

REGRESSION IN R

PEARSON CORRELATION

Regression Analysis

Simple Linear Regression

REFER to:
lectureSEVEN #02

EXAMPLE (slide 19)

Example

House Price in \$1000s (y)	Square Feet (x)
245	1400
312	1600
279	1700
308	1875
199	1100
219	1550
405	2350
324	2450
319	1425
255	1700

Regression Analysis

Simple Linear Regression

- To perform linear regression in R, use the **lm()** function
- This will build a **linear regression model** based on the data
- How to use the function:
 - Assume you have your x, y
 - Use: **lm(y~x)**
 - Expression of “~” in R means it is a formula. Where y~x is equivalent to $y = f(x)$

```
1  x <- c(1400, 1600, 1700, 1875,  
2  1100, 1550, 2350, 2450, 1425,  
3  1700)  
4  
5  y <- c(245, 312, 279, 308, 199,  
6  219, 405, 324, 319, 255)  
7  
8  model <- lm(y~x)  
9  model  
10  
11 summary(model)  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22
```

Linear regression model

$$\hat{y} = 98.2483 + 0.1098x$$

Regression Analysis

Simple Linear Regression

- To perform linear regression in R, use the **lm()** function
- This will build a **linear regression model** based on the data
- How to use the function:
 - Assume you have your x, y
 - Use: **lm(y~x)**
 - Expression of “~” in R means it is a formula. Where y~x is equivalent to $y = f(x)$

```

1  x <- c(1400, 1600, 1700, 1875,
2  1100, 1550, 2350, 2450, 1425,
3  1700)
4
5  y <- c(245, 312, 279, 308, 199,
6  219, 405, 324, 319, 255)
7
8  model <- lm(y~x)
9  model

```

```

> model

Call:
lm(formula = y ~ x)

Coefficients:
(Intercept)          x
  98.2483       0.1098

```

Linear regression model

$$\hat{y} = 98.2483 + 0.1098x$$

Regression Analysis

Simple Linear Regression

Try adjust the plot with custom limits for x-axis and y-axis

```
> summary(model)
```

```
Call:
lm(formula = y ~ x)
```

$$\hat{y} = 98.2483 + 0.1098x$$

```
Residuals:
```

Min	1Q	Median	3Q	Max
-49.388	-27.388	-6.388	29.577	64.333

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	98.24833	58.03348	1.693	0.1289
x	0.10977	0.03297	3.329	0.0104 *

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 41.33 on 8 degrees of freedom
```

```
Multiple R-squared:  0.5808,    Adjusted R-squared:  0.5284
```

```
F-statistic: 11.08 on 1 and 8 DF,  p-value: 0.01039
```

Regression Analysis

Simple Linear Regression

- What if we want to see the plot of the model we learn from the data?
- First, plot a scatter plot based on x and y using **plot()** function
- Add the linear regression model into the plot using **abline()** function

```
1  x <- c(1400, 1600, 1700, 1875,  
2  1100, 1550, 2350, 2450, 1425,  
3  1700)  
4  
5  y <- c(245, 312, 279, 308, 199,  
6  219, 405, 324, 319, 255)  
7  
8  model <- lm(y~x)  
9  model  
10  
11 plot(x,y)  
12 abline(model)  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22
```

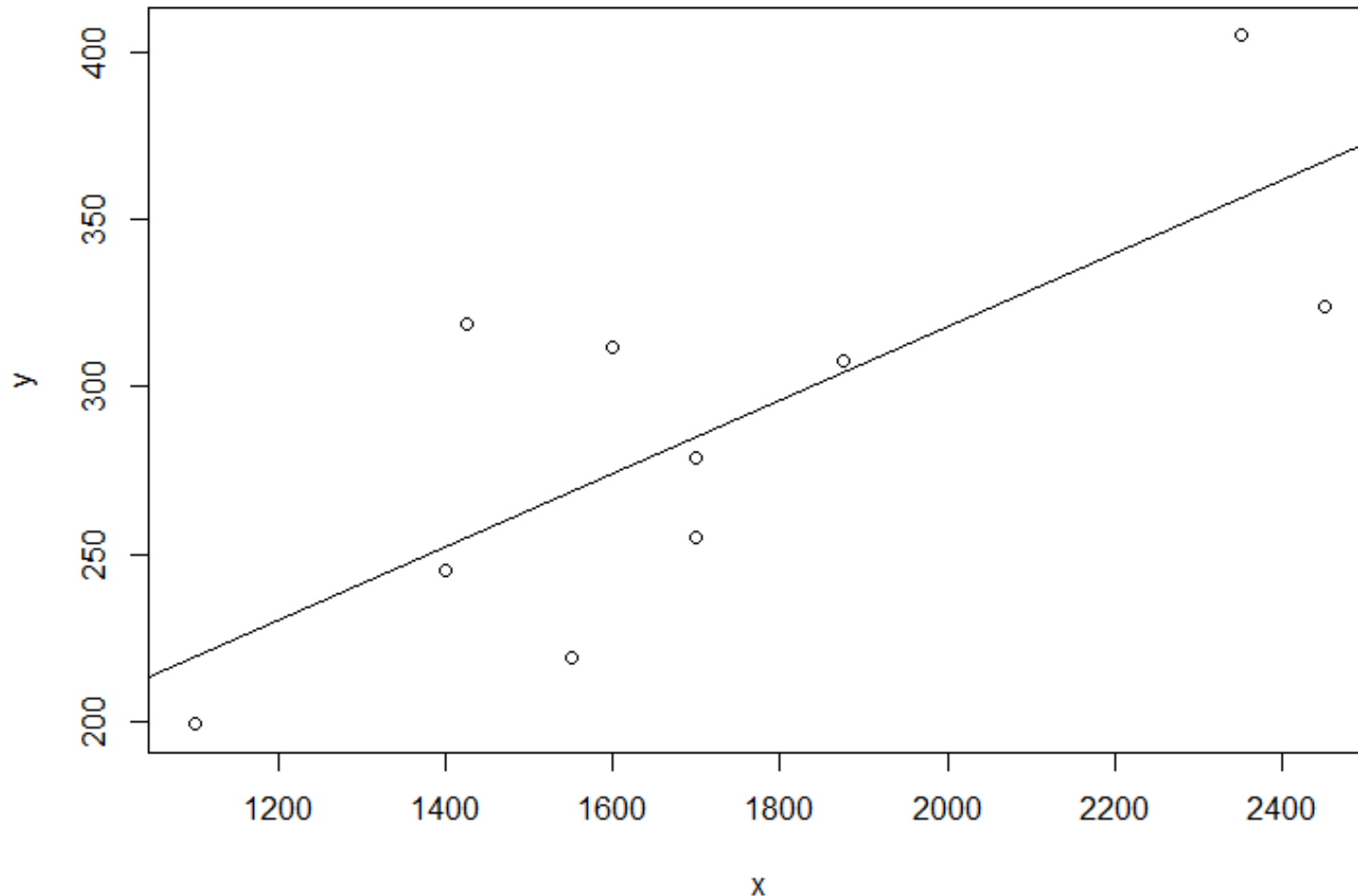
Linear regression model

$$\hat{y} = 98.2483 + 0.1098x$$

Regression Analysis

Simple Linear Regression

Try adjust the plot with custom limits for x-axis and y-axis



Regression Analysis

Simple Linear Regression

Replotting using:
`plot(x,y, xlim=c(0,2600), ylim=c(0,500))`
`abline(model)`

